

SORTING

Q8)

Question:

DS	Session	Sorting	Question Information	Level 1 Challenge 18
Question description kkalaiselvan is going to behave as a vehicle driver. he needs to drive an automobile on a track divided into "N" no. of sub-tracks. The letter "K" is also assigned to you. i.e. the maximum number of kilometres a vehicle may go on each sub-track. You can add any unit of Petrol to the automobile if it can't cover a sub-track. The total kilometres your automobile may go increases by one unit for every unit of gasoline added. Input: The first line of input contains an integer T denoting the no of test cases. Then T test cases follow. Each test case contains two space separated integers N and K. The second line of each test case contains N space separated integers (A[]) denoting the distance of each N sub-tracks. Output: For each test case in a new line you have to print out the minimum unit of Petrol your car require to cover all the sub-tracks. If no extra unit of petrol is required, print -1.				

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int T, N, K, i, j, temp;
5
6      scanf("%d", &T);
7
8      while (T--) {
9          scanf("%d %d", &N, &K);
10
11         int A[N];
12
13         for (i = 0; i < N; i++)
14             scanf("%d", &A[i]);
15
16         for (i = 0; i < N - 1; i++) {
17             for (j = i + 1; j < N; j++) {
18                 if (A[i] > A[j]) {
19                     temp = A[i];
20                     A[i] = A[j];
21                     A[j] = temp;
22                 }
23             }
24         }
25
26         if (A[N - 1] <= K)
27             printf("-1\n");
28         else
29             printf("%d\n", A[N - 1] - K);
30     }
31
32     return 0;
33 }
```

OUTPUT:

TEST CASE 1:

```
2
5 6
2 5 4 5 2
-1
5 3
1 6 3 5 2
3
```

TEST CASE 2:

```
2
5 6
12 15 14 15 12
9
5 3
11 61 31 51 21
58
```